

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402672
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Zhang; Lei et al.

Reusable protective clothing

Abstract

A reusable protective clothing includes a one-piece protective suit which includes integrated top and trousers attached with gloves and shoe covers, and a protective headpiece having a face shield. A front side of the protective headpiece is provided with a full-vision goggle with 180-degree field of view, the full-vision goggle is integrated with the face shield, a back side of the protective headpiece is provided with a protective hood made of a material of the one-piece protective suit, the protective hood is integrated with the top, the front side of the protective headpiece is provided with an air vent for a mouth and a nose configured according to facial features of a wearer, and an outer side of the protective headpiece is provided with ear-shaped hooks. The protective clothing is designed according to application environment and medical requirements to prevent exposure to pathogenic microorganisms.

Inventors: Zhang; Lei (Xiamen, CN), Shao; Yuzhang (Xiamen, CN), Zhao; Dean (Xiamen, CN), Li; Ye (Xiamen, CN), Wen; Leji (Xiamen, CN), Lin; Xin (Xiamen, CN)

Applicant: XIAMEN UNIVERSITY (Xiamen, CN)

Family ID: 84989625

Assignee: XIAMEN UNIVERSITY (Xiamen, CN)

Appl. No.: 18/063046

Filed: December 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240130448 A1	Apr. 25, 2024
US 20240237763 A9	Jul. 18, 2024

Foreign Application Priority Data

CN	202211279309.X	Oct. 19, 2022
----	----------------	---------------

Publication Classification

Int. Cl.: A41D13/02 (20060101); A41D13/11 (20060101)

U.S. Cl.:

CPC A41D13/02 (20130101); A41D13/1184 (20130101); A41D2200/20 (20130101)

Field of Classification Search

CPC: A41D (13/02); A41D (13/1184); A41D (13/1218); A41D (13/1161); A31D (2200/20); A31D (2500/20); A62B (17/006)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4253198	12/1980	Estabrook	2/84	B63C 9/087
4845779	12/1988	Wheeler	2/84	A42B 1/046
5918314	12/1998	Moses	2/84	A41D 13/02
11202925	12/2020	Awad	N/A	A62B 7/10
2005/0204451	12/2004	Plut	2/161.6	A41D 19/0055
2015/0113716	12/2014	Samaniego	2/457	A42B 1/048
2021/0267292	12/2020	Wang	N/A	A41D 13/12
2021/0298368	12/2020	Bussey	N/A	A47C 31/003
2022/0110387	12/2021	Detorre	N/A	A41D 13/1184
2022/0160059	12/2021	Arangdad	N/A	A41D 31/10

OTHER PUBLICATIONS

“Nanomaterial sticky tape that withstands extreme temperatures”, Laurie Donaldson, Materials Today, Jul. 30, 2019, accessed from <https://www.materialstoday.com/nanomaterials/news/nanomaterial-sticky-tape-extreme-temperatures/> (Year: 2019). cited by examiner

Primary Examiner: Lynch; Patrick J.

Assistant Examiner: Duckworth; Brianna T.

Attorney, Agent or Firm: JEEN IP LAW, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims priority to Chinese Patent Application No. 202211279309.X, filed on Oct. 19, 2022, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of protective clothing, particularly to a reusable protective clothing.

BACKGROUND

(3) At present, the majority of protective clothing consumed globally is disposable protective clothing, which has the following drawbacks: (1) Substantial environmental pollution: the extensive use of disposable protective clothing requires high disposal costs and causes substantial environmental pollution. (2) Poor physical strength: the currently available disposable protective clothing has poor mechanical properties and tends to be easily torn by an external force, thus it cannot provide adequate protection. Most of the reusable protective clothing developed worldwide use polytetrafluoroethylene (PTFE) film with micropores or PTFE composite film to form a barrier layer. In order to improve the laundering durability, a thick barrier film is preferred, which also has the following drawbacks. (1) The protective clothing needs to be supplemented with goggles, a face shield, a mask, gloves, and shoe covers for full body protection, and therefore cannot be put on or taken off as a whole. (2) Uncomfortable to wear: its key barrier layer features poor moisture permeability and can impair visual field and voice transmission. (3) Lifespan, namely the number of repeated use, of the protective clothing is very limited. (4) There is no differentiation between its use in the medical field and epidemic prevention.

(4) Currently, due to the continuous waves of the pandemic, as well as the increase in awareness of disease prevention and volunteering, more and more people participate in disease prevention and control. The reusable protective clothing of the present disclosure is specially designed for disease prevention personnel and indoor medical staff. On the one hand, the reusable protective clothing can reduce medical waste and save social resources. On the other hand, the reusable protective clothing can reduce the number of steps for personnel to put on and take off the protective clothing, so as to maximize the protection for non-professionals against exposure to pathogens caused by operational mistakes. In addition, auxiliary functions are added to the reusable protective clothing in consideration of the specific requirements of disease prevention personnel and medical staff.

SUMMARY

(5) The present disclosure provides a reusable protective clothing which can effectively address the above problems.

(6) The present disclosure provides a reusable protective clothing material including a high-density woven fabric layer, a middle-layer fabric and an outer-layer fabric.

(7) A reusable protective clothing is made of the reusable protective clothing material.

(8) The reusable protective clothing is a one-piece protective suit including integrated top and trousers attached with gloves and shoe covers, and further including a protective headpiece having a face shield, wherein a front side of the protective headpiece is provided with a pair of full-vision goggles with 180-degree field of view, the pair of full-vision goggles is integrated with the face shield, a back side of the protective headpiece is provided with a protective hood made of a material of the one-piece protective suit, the protective hood is integrated with the top, the front side of the protective headpiece is also provided with an air vent for a mouth and a nose configured according to facial features of a wearer, and an outer side of the protective headpiece is provided with ear-shaped hooks.

(9) As a further improvement, the air vent for the mouth and the nose is arranged on the protective headpiece at a position corresponding to the mouth and a bridge of the nose of the wearer, and an edge of the air vent is provided with a sealing ring for fixing a filter membrane.

(10) As a further improvement, an elastic belt is arranged along an edge around an outer contour of the filter membrane.

(11) As a further improvement, the protective headpiece further includes a latex band arranged on two sides of the air vent for the mouth and the nose, and the latex band is arranged inside the protective headpiece to assist the positioning of the air vent for the mouth and the nose.

(12) As a further improvement, the protective headpiece further includes a support ring arranged at

a top part of the protective headpiece.

(13) As a further improvement, the support ring is a semicircular silicone ring with a thickness of 3 cm.

(14) As a further improvement, the top includes a back body covering piece and a front body covering piece, the protective hood is connected to a neck position of the back body covering piece and extends to the top part of the protective headpiece, and a first end of a waterproof zipper is arranged at a hip position at a back side of the trousers, a second end of the waterproof zipper extends through the back body covering piece to the top of the protective hood, and the waterproof zipper is covered with a flap.

(15) As a further improvement, the gloves are full-finger wear-resistant anti-slip gloves.

(16) As a further improvement, the gloves are half-finger gloves, and a wrist position of each half-finger glove is provided with nano tape.

(17) As a further improvement, the trousers further include wear-resistant and anti-slip shoe covers provided at bottom positions of trouser legs.

(18) The present disclosure has the following advantages. The present disclosure provides a reusable one-piece protective clothing designed according to the needs of epidemic prevention personnel and volunteers, and provides a medical protective clothing designed according to the needs of medical personnel. The protective clothing is a one-piece protective suit that is designed according to the environment and medical requirements and can maximize prevention against exposure to pathogenic microorganisms during wearing, taking-off and using process. The term "one-piece" reflects in the design that only the nose and the mouth are accessible from the outside, and the rest parts of the body are fully enclosed as a closed loop. The protective clothing is lightweight and requires no additional auxiliary device or means. Also, the protective clothing is able to be put on and taken off as a whole and thus the putting-on and taking-off can both be accomplished by a single act.

(19) The medical protective clothing and the reusable one-piece protective clothing both have a protective headpiece with a face shield. Namely, the front side of the protective headpiece is a 180-degree fully transparent head cover made of glass fibre-reinforced plastic material, and the back side of the protective headpiece is a protective hood made of the material of the protective clothing. This design not only increases the field of view, but also minimizes hearing loss by air transmission, it has better performance than the design of the head covers of known protective clothing on the market and is easier to use.

(20) The filter membrane is specially designed for the mouth and the nose, and thus a conventional face mask may not be worn, thereby reducing the strain on the ears. Meanwhile, the filter membrane for the mouth and the nose serves as an access for ventilation of the protective headpiece as well as providing protection for the wearer, which is favorable for permeation of exhaled moisture, and reducing moisture vapor inside the protective headpiece. Furthermore, ear-shaped hooks are provided on an outer side of the protective headpiece, so that an additional mask may be worn on the hooks in special settings to provide dual-layer protection for the respiratory tract.

(21) The present disclosure is configurable with full-finger wear-resistant anti-slip gloves or half-finger gloves according to practical requirements. For example, half-finger gloves may be used indoors, and the fingers are covered with medical rubber gloves, so as to improve the flexibility of the fingers and allow the medical personnel to do some fine finger movements and operations. Alternatively, the full-finger wear-resistant anti-slip gloves may be used outdoors or in some special settings. The fully enclosed design provides a higher protection level. Furthermore, friction particles provided on the gloves offer increased friction to prevent items from slipping off the wearer's grip.

(22) According to the work practices of the epidemic prevention personnel and volunteers, the elbow and knee positions of the protective clothing are provided with reflective fabric pieces to

improve the wear resistance of the quick-wearing parts and to increase the safety factor of outdoor work at night.

(23) The material of the reusable protective clothing can withstand immersion in chlorine-containing disinfectant with a concentration of 1000 mg/l for 60 minutes, and more than 50 times of disinfection. The material of the one-piece medical protective clothing can withstand moist heat sterilization under a temperature of 115° C.-121° C. for 15 min-30 min, and more than 50 times of moist heat sterilization, thereby permitting the reuse of the protective clothing several times and reducing the generation of medical waste.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to make the technical solutions of the embodiments of the present disclosure clearer, the drawings used to illustrate the embodiments will be briefly introduced below. It should be understood that the following drawings merely show some embodiments of the present disclosure, and therefore should not be regarded as restriction to the scope of protection. For those of ordinary skill in the art, other related drawings may be derived based on these drawings without any creative effort.

(2) FIG. 1 is a schematic diagram showing a front side of a reusable protective clothing according to a first embodiment of the present disclosure.

(3) FIG. 2 is a schematic diagram showing a back side of a reusable protective clothing according to the first embodiment of the present disclosure.

(4) FIG. 3 is a schematic diagram showing a front side of a protective headpiece having a face shield of the reusable protective clothing according to an embodiment of the present disclosure.

(5) FIG. 4 is a schematic diagram showing a back side of a protective headpiece having the face shield of the reusable protective clothing according to an embodiment of the present disclosure.

(6) FIG. 5 is a schematic diagram showing a structure of an air vent for the mouth and the nose of the protective headpiece having the face shield of the reusable protective clothing according to an embodiment of the present disclosure.

(7) FIG. 6 is a schematic diagram showing a structure of a filter membrane for the mouth and the nose of the reusable protective clothing according to an embodiment of the present disclosure.

(8) FIG. 7 is another schematic diagram showing a structure of a filter membrane for the mouth and the nose of the reusable protective clothing according to an embodiment of the present disclosure.

(9) FIG. 8 is a schematic diagram showing a front side of a reusable protective clothing according to a second embodiment of the present disclosure.

(10) FIG. 9 is a schematic diagram showing a back side of the reusable protective clothing according to a second embodiment of the present disclosure.

(11) FIG. 10 is a schematic diagram showing the fabric layers of the material of the reusable protective clothing according to the first and second embodiment of the present disclosure.

(12) FIG. 11 shows the bacteriostatic effect of the medical protective clothing of the reusable protective clothing using chitosan antibacterial agent with different mass concentrations according to the first embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(13) In order to make the objectives, technical solutions, and advantages of the embodiments of the present disclosure clearer, the technical solutions of the embodiments of the present disclosure will be described clearly and completely below in combination of the drawings used to illustrate the embodiments of the present disclosure. Obviously, the described embodiments are parts of rather than all of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments derived by those of ordinary skills in the art without creative

effort fall within the scope of protection of the present disclosure. Accordingly, the following detailed description of the embodiments of the present disclosure shown in the drawings is not intended to limit the scope of protection of the present disclosure, but merely to represent selected embodiments of the present disclosure.

(14) In the description of the present disclosure, the terms “first”, “second” are used only for descriptive purposes and cannot be understood as indicating or implying relative importance or implying the number of indicated technical features. Thus, a characteristic that is referred to by “first” and “second” may include, expressly or implicitly, one or more of the characteristics. In the description of the present disclosure, “multiple/several/couple” means two or more, unless otherwise expressly specified.

Embodiment 1

(15) Referring to FIGS. 1-7, the present embodiment provides a specific application of the reusable protective clothing, i.e. medical protective clothing, which is used indoors or at some places with relatively flat ground, such as hospitals, clinics, etc., and is mainly involved in indoor activities, such as medical treatment activities, which requires fine finger movements and operations. The protective clothing is mainly used to protect the wearer against all pathogenic microorganisms such as viruses, bacteria, rickettsia, and so on from internal and external environments. The medical protective clothing mainly includes a one-piece protective suit **10** and a protective headpiece **20** having a face shield **11**.

(16) The one-piece protective suit **10** includes integrated top **101** and trousers **102**, and the top **101** and trousers **102** are respectively attached with gloves and shoe covers. In practice, the wearer may put on the one-piece protective suit **10** by: first wearing the trousers **102**, then wearing the top **101**, afterwards wearing the protective headpiece **20** with the face shield **11**, and finally pulling up the waterproof zipper **1014** with the help of another individual to seal the protective suit. The specific steps and principles are described as follows.

(17) **S1**, wearing the trousers **102**.

(18) As shown in FIG. 1 and FIG. 2, the trousers **102** of the present disclosure further include wear-resistant and anti-slip shoe covers **1021** connected to the bottom of the trouser legs. The soles of the shoe covers of the known protective clothing are also made of the same material as the top **101** and the trousers **102**, so the soles may easily be damaged. According to the present disclosure, instead of being made of the material of the protective clothing, the wear-resistant and anti-slip shoe covers **1021** are made of another material to increase the thickness of the bottom, so as to allow medical staff to wear the shoe covers with their own shoes on while the wear-resistant and anti-slip shoe covers **1021** are unlikely to be damaged during walking.

(19) The wear-resistant and anti-slip shoe covers **1021** of the present disclosure are made of rubber material which has many advantages including good wear resistance, anti-slip performance, elasticity, anti-break strength, suppleness, extensibility, stable shrinkage, hardness, flexibility and so on. These advantages allow the shoe covers to withstand repeated sterilization and repeated use in practice.

(20) Further, the thickness of the sole of the wear-resistant and anti-slip shoe cover **1021** is 1-2 cm.

(21) In one specific embodiment, the thickness of the sole of the wear-resistant and anti-slip shoe cover **1021** is 1.5 cm which ensures that the wear-resistant and anti-slip shoe cover **1021** will not be damaged by walking while the sole is light-weight and not too thick to affect the comfort degree of walking. The thickness of the sole plus the thickness of the wear-resistant and non-slip shoe cover **1021** is smaller than or equals to 3 cm.

(22) Further, the sole of the wear-resistant and anti-slip shoe cover **1021** is provided with friction patterns which include but not limited to wavy patterns. In indoor settings such as hospitals etc., the floor is frequently mopped and washed in these places as required by medical standards, so the ground is often wet and slippery. The design of the friction pattern on the bottom side of the sole increases friction between the said bottom side and the ground, so the wearer is unlikely to slip and

fall off during walk.

(23) In one specific embodiment, the friction pattern is wavy-shaped with a depth of 0.1-0.5 cm. The friction pattern being a wavy pattern with a depth of 0.15 cm can achieve an optimal friction resistance with the ground, and the pattern is unlikely to be substantially worn in a single use even after prolonged walking, so as to achieve repeated use.

(24) In one specific embodiment, the friction pattern is circular-shaped with a depth of 0.15-0.4 cm. The friction pattern being circular-shaped with a depth of 0.2 m and a diameter of 2.5 cm can achieve an optimal friction resistance with the ground, and the pattern is unlikely to be substantially worn in a single use after prolonged walking, so as to achieve the effect of repeated use.

(25) As a further improvement, positions of the ankle of the wear-resistant and anti-slip shoe cover **1021**, the wrist position, and waist position are provided with hook and loop fasteners **1022** for fastening and fixing the protective clothing.

(26) S2, wearing the top **101**.

(27) As shown in FIG. 1 and FIG. 2, a pair of medical rubber gloves can be put on the wearer's hands before putting on the top **101**, and after the top **101** is put on another pair of medical rubber gloves are put on the wearer's hands.

(28) The top **101** includes a back body covering piece **1011** and a front body covering piece **1012**, a protective hood **1013** is connected to a neck position of the back body covering piece **1011** and extends to the top part of the protective headpiece **20**, and a first end of a waterproof zipper **1014** is arranged at a hip position at a back side of the trousers **102**, a second end of the waterproof zipper **1014** extends through the back body covering piece **1011** to the top of the protective hood **20**, and the waterproof zipper **1014** is covered with a flap **1015**.

(29) As shown in FIG. 1 and FIG. 2, the gloves are half-finger gloves **1016**. After wearing the half-finger gloves **1016**, a pair of medical rubber gloves are put on. The half-finger gloves **1016** allow the fingers to move flexibly for conducting fine finger movements and operations, such as operating surgical instruments and so on. Since the fabric material of the one-piece protective suit **10** is relatively rough and stiff, the use of half-finger gloves **1016** can maximize the flexibility of the fingers and improve the sealability of the medical protective clothing. Since medical rubber gloves are worn before and after wearing the protective clothing, two layers of medical rubber gloves ensure that even if the outer layer of medical rubber gloves is taken off after use, the air tightness of the protective clothing will not be affected, and the inner layer of medical rubber gloves will not be contaminated, so as to achieve the purpose of repeated use.

(30) As shown in FIG. 1 and FIG. 2, a nano-tape **1017** is provided at the wrist position of the half-finger gloves **1016**. The nano-tape **1017** is used to fix the rim of the medical rubber gloves to the wrist position of the medical protective clothing. Further, if the damage of the protective clothing is discovered during use, the nano-tape **1017** can be used to patch the tear to temporarily repair the protective clothing.

(31) The nano-tape **1017** is made of nano-adhesive material and can withstand multiple times of washing. When the adhesive surface of the nano-tape is contaminated by dust or other dirt, the nano-tape can be washed with clean water and dried while retaining its original adhesiveness. Also, the nano-tape is removable without leaving any residue, so it can be applied for attachment and removed repeatedly without leaving any residue on the surface of objects.

(32) Further, the width of the nano-tape **1017** is 4-6 cm. In one specific embodiment, the width of the nano-tape **1017** is 5 cm, which is an optimal width because, given the limited length usually used for patching, the tape will not get folded up transversely onto itself during use, so the nano-tape is simpler and more convenient for use.

(33) Further, a pocket is provided at the position of the chest on the front body covering piece **1012**, and nano-tape **1017** is placed in the pocket as a backup.

(34) As a further improvement, the upper arm position of the top **101** is provided with an armband attachment **1018** for attaching an identification tag of the wearer, and the identity information

carried on the identification tag includes but not limited to the serial number and name, so as to facilitate the identification of the medical staff.

(35) As a further improvement, absorbent bags **1019** may be provided and fixed at multiple places, such as the center position of the chest on the front body covering piece **1012**, the center position of the back, and the underarm positions, by means of hook and loop. The absorbent bags **1019** can absorb the moisture from corresponding locations covered by the one-piece protective suit **10**. For example, the absorbent bag **1019** is arranged at the center of the chest where body fluid is most likely to be produced from the surface of the wearer's body and along which body fluid from the neck flows down, such that the body fluid can be directly absorbed by the absorbent bag **1019**. Since the absorbent bag **1019** is detachable, the absorbent bag **1019** can be removed before disinfection and sterilization.

(36) Furthermore, one or more hook and loop fasteners may be arranged between the absorbent bags **1019** and the inner wall of the protective clothing according to the body shape of the wearer or other factors. Compared with the configuration of one hook and loop fastener, multiple loop and hook fasteners provided at different connection points enable the absorbent bags **1019** to be attached more securely, and can prevent detachment of the heavier absorbent bag **1019** after absorbing body fluid.

(37) In practice, compared with directly wearing the protective clothing, placing the absorbing bag **1019** can improve the comfortableness of the protective clothing as the absorbent bag **1019** can absorb body fluid produced by wearing the protective clothing under static or dynamic working conditions.

(38) Preferably, CaCl.sub.2 may be placed in the absorbent bag **1019**. Compared with other ordinary materials, the absorbent bag **1019** with CaCl.sub.2 has better absorbing performance.

(39) More preferably, molecular sieve may be placed in the absorbing bag **1019**. Compared with CaCl.sub.2, molecular sieve further improves the absorbing performance of the absorbing bag **1019**.

(40) **S3**, wearing a protective headpiece **20** having a face shield **11**.

(41) With reference to FIG. 3-FIG. 7, the present disclosure further includes a protective headpiece **20** having a face shield **11**. The front side of the protective headpiece **20** is provided with a pair of 180-degree full-vision goggles **12** which is integrated with the face shield **11**. The back side of the protective headpiece **20** is provided with a protective hood **1013** made of protective clothing material, and the protective hood **1013** is integrated with the top **101**. The front side of the protective headpiece **20** is provided with an air vent **201** for the mouth and the nose corresponding to the facial features of the wearer. Compared with the prior art in which only a pair of goggles is used for viewing in a limited visual angle, the present disclosure enables the wearer to view in a wider visual angle and is more practical to use. Here, the connection method of the protective headpiece **20** and the protective hood **1013** includes but not limited to press fit, sheathing, etc., and the protective headpiece **20** can be removed and repeatedly used in other protective clothing.

(42) As shown in FIG. 5, the protective headpiece **20** is provided with a silicone sealing ring **203** at the edge of the air vent **201** for the mouth and the nose corresponding to the mouth and bridge of the nose of the wearer. The use of flexible material such as silicone can ensure a better fit with the shape of the wearer's face without leaving a gap, thereby improving the air tightness of the medical protective clothing.

(43) Further, a U-shaped groove is formed between the sealing ring **203** and the inner wall of the protective headpiece **20**, and the filter membrane **205** for the mouth and the nose is fixed at the U-shaped groove with the assistance of a latex band **204**, so as to eliminate the need of a mask, and thus reduces the strain on ears. The use of the special filter membrane for the mouth and the nose is favorable for discharging moisture vapor from the wearer's respiratory tract, and the specific principle is described as follows. Compared with the completely enclosed protective headpiece with a face shield in the prior art, since there is a temperature difference between the air exhaled

from the respiratory tract and the protective headpiece, the moisture vapor tends to condense into water droplets on the protective headpiece. Epidemic prevention and medical staffs who work in this environment for a prolonged period get sweaty easily, leaving their body and clothes wet. According to the present disclosure, filter membrane **205** is used as an access to connect the inside and outside of the protective headpiece **20** for protection and ventilation, thereby further reducing moisture generation in the protective headpiece **20**.

(44) Further, the thickness of the side of the sealing ring **203** in contact with the inner wall of the protective headpiece **20** is greater than that of the opposite side. The side of the sealing ring has a lower thickness which enables it to better fit with the wearer's face and is unlikely to detach therefrom.

(45) Further, the sealing ring **203** forms an angle with the horizontal plane, and the angle ranges 90° - 150° . In one specific embodiment, the angle between the sealing ring **203** and the horizontal plane is 100° . With a relatively smaller tilt angle, the sealing ring **203** can be easily attached in place with wearer's mouth and nose without the need for repeated adjustments, and thus the air vent **201** can be directly aligned with the facial features as the sealing ring is attached to the wearer's face.

(46) Further, the structure of the filter membrane **205** for the mouth and the nose shown here includes two layers of fibrous non-woven fabrics for professional medical and health care use and a middle layer between the two layers. The middle layer is made of filter melt-blown fabric which can filter and prevent passage of more than 99% of bacteria. The middle layer and the other two layers are combined by ultrasonic welding.

(47) As shown in FIG. 7, further, an elastic belt **2051** is arranged along an edge around an outer contour of the filter membrane **205** for the mouth and the nose. When fixing the filter membrane **205** on the sealing ring **203**, the elastic belt **2051** is adaptable to the shape of the air vent **201** for the mouth and the nose and tightly engaged with the U-shaped groove of the sealing ring **203**, so as to prevent any displacement with the movement of the mouth, and further improve the air tightness of the protective clothing. In addition, the filter membrane **205** can be handled as medical waste after use.

(48) The protective headpiece **20** of the present disclosure further includes ear-shaped hooks **202** provided on the outer side of the protective headpiece **20**. If enhanced protection for the respiratory tract is required under special circumstances, a mask may be worn on the external ear-shaped hooks **202** to provide a secondary filtration of air and forms a dual-layer protection for the respiratory tract in combination with the filter membrane **205** for the mouth and the nose.

(49) The protective headpiece **20** of the present disclosure further includes a latex band **204** arranged on two sides of the air vent **201**, and the latex band **204** is arranged inside the protective headpiece **20** to help the positioning of the air vent **201**.

(50) The protective headpiece **20** of the present disclosure further includes a support ring **206** arranged at a top part of the protective headpiece **20** to stably support the head part of the protective headpiece **20**, and to retain its shape and avoid deformation.

(51) Further, the support ring **206** is made of a semi-circular silicone ring with a thickness of 3 cm, and the support ring **206** supports the top part of the protective headpiece **20**. After the protective headpiece **20** is put on, the inner wall of the protective headpiece **20** is supported and will not attach to the wearer's forehead.

(52) Currently available fully enclosed protective headpiece with a face shield is generally a fully integrated helmet, which completely covers the entire head and can easily cause sound loss. Namely, by using a fully enclosed headpiece, the wearer cannot be heard clearly and the wearer also cannot hear others clearly.

(53) According to the present disclosure, the front side of the protective headpiece **20** is provided with a 180-degree fully transparent head cover made of glass fiber-reinforced plastic, and the back side of the protective headpiece **20** is provided with a protective hood **1013** made of the material of

the protective clothing. In order to ensure a barrier-free 180-degree field of view and minimize air transmission loss, hearing comparison tests under different settings are performed during the design of the protective headpiece **20**.

(54) First, sound vibrations are transmitted to the auditory center of the brain to create hearing by two types of conduction. One of them is air transmission where sound vibrations are transmitted to the eardrum through external auditory canal, then transmitted to the cochlear hair cells through the ossicular chain to generate electrical signals which are then transmitted to the auditory center along the auditory nerve. Another one is bone conduction, where sound vibrations are transmitted to the cochlear hair cells through the skull bone, and follow the same pathway as the air transmission to be transmitted to the auditory center.

(55) Accordingly, the present disclosure performed sound field tests, the principle is that the hearing of the tested person is determined according to the response of the person to the quietest sound that can be heard. The sound field tests were carried out by the following method. In the same test environment, the brand of audiometer used was OTOMETRICS® and the brand of speaker used was SOUNDLINK®, the hearing of the volunteers was tested in 3 conditions including: baseline hearing of the volunteers themselves in normal situations, hearing after wearing the protective clothing, and hearing after wearing the head cover. The audiometer emits sound waves of different frequencies to be transmitted into the sound field through the speaker, and the volunteers response to whether they hear the sound or not to determine the hearing threshold.

(56) Circumstance of statistical significance $P>0.05$ was not found in the statistical results under normal hearing conditions, which indicates that the overall distribution of the samples is non-skewed, and the data can be regarded as satisfying normality, so paired samples t-test was performed for pairwise comparisons.

(57) When the difference between wearer's own baseline hearing and the wearer's hearing with the protective clothing on under the normal situation is not statistically significant. i.e. $P>0.05$, it can be considered that the protective clothing will not cause hearing loss; when the difference between the wearer's own baseline hearing and the wearer's hearing with the protective head cover on under normal situation is statistically significant, i.e. $P<0.01$, which indicates that the overall distribution of the two samples is different, it can be considered that the protective head cover will cause significant hearing loss; when the difference between the hearing with the protective clothing on and the hearing with the protective head cover on is statistically significant, i.e. $P<0.01$, which indicates that the overall distribution of the two samples is different, it can be considered that the protective head cover will cause significant hearing loss compared with the protective clothing.

(58) TABLE-US-00001 TABLE 1 Paired sample test of audiometry under three situations paired difference Difference 95% standard confidence interval average standard error lower upper degrees of Sig. value deviation mean limit limit t freedom Two-tailed pair 1 Normal- 0.45455 3.50325 1.05627 -1.89897 2.80806 0.430 10 0.676 protective clothing pair 2 Normal- -13.29545 3.12614 0.94257 -15.39562 -11.19529 -14.106 10 0.000 headpiece pair 3 Protective -13.75000 3.06186 0.92319 -15.80699 -11.69301 -14.894 10 0.000 clothing- headpiece

(59) To sum up, this experiment determined the level of sound according to the volunteers' response to sound in the three situations to confirm the degree of sound loss of air transmission caused by the protective clothing and head cover. The results showed that the difference between hearing loss with the protective clothing on and hearing loss in the normal state is not statistically significant. While with the protective head cover on, whether compared with the protective clothing test or normal state test, it all shows that hearing loss is significant due to the limited air transmission. Therefore, in order to ensure the barrier-free 180-degree field of view of the protective clothing of the present disclosure, the head cover designed by the present disclosure includes a front part and a back part. The front part is a protective headpiece **20** with a face shield **11** and is made of glass fiber-reinforced plastic material to ensure a full-screen field of view, and the back part of the head cover is made of the protective clothing material to minimize air

transmission loss, and the improved head cover of the protective clothing has better performance than those are currently available on the market and is easier to use.

(60) **S4**, pull up the waterproof zipper **1014** with the help of another person, and put the flap **1015** in place for covering the waterproof zipper **1014**, thus the putting-on and taking-off can both be accomplished by a single act.

Embodiment 2

(61) As shown in FIGS. **8** and **9**, the present embodiment is different from embodiment 1 in that the present embodiment relates to a reusable protective clothing for epidemic prevention, which is mainly worn and used outdoors, including but not limited to forests, urban streets, grasslands, sandy landscapes, etc., where social activities take place and there are labor-intensive operations, the object to be prevented against mainly includes internal and external environmental viruses.

(62) Structurally, this embodiment has the following differences from embodiment 1.

(63) Above all, there are differences in the structure of the one-piece protective suit **10**.

(64) First, the trousers **102** further include wear-resistant and anti-slip shoe covers **1021** connected to the bottom of the trouser legs. Here, the sole of the wear-resistant and anti-slip shoe covers **1021** have a thickness of 2-3.5 cm. By increasing the thickness of the sole of the shoe covers, the friction between the sole and the ground when walking outdoors is improved and no substantial wear will be sustained after one use, so as to achieve the effect of repeated use.

(65) In one specific embodiment, the friction pattern is wavy with a depth of 0.2-0.5 cm. The friction pattern being a wavy pattern with a depth of 0.3 cm can get an optimal friction resistance with the ground, and the pattern is unlikely to be substantially worn in a single use after walking for a prolonged period, so as to achieve the repeated use.

(66) In one specific embodiment, the friction pattern is circular-shaped with a depth of 0.15-0.4 cm. The friction pattern being circular-shaped with a depth of 0.3 m and a diameter of 2.5 cm can get an optimal friction resistance with the ground, and the pattern is unlikely to be substantially worn in a single use after walking for a prolonged period so as to achieve the repeated use.

(67) Second, the knee and elbow positions of the trousers **102** and top **101** are covered with reflective fabrics **103** to ensure the safety of medical personnel to some extent. On one hand, the reflective fabric can improve the wear resistance of the quick-wearing parts. On the other hand, the reflective fabric can increase the safety factor of outdoor work at night. Even if the medical staff works outdoors, especially at night, the reflective fabric can reflect incident light and the reflected light helps others to locate the medical staff.

(68) The methods to combine the reflective fabrics here include, but not limited to, glue bonding, thermoforming, and hook and loop fasteners and so on, as long as the surface of the one-piece protective suit **10** will not be damaged.

(69) Third, the gloves are full-finger wear-resistant and non-slip gloves **1020**. Compared with the half-finger gloves **1016** used in combination with medical protection gloves, the protective clothing in the embodiment is mainly used outdoors, while medical rubber gloves are prone to wear and damage and are not easily repairable. The full-finger wear-resistant anti-slip gloves **1020** have a rough surface and are not easy to be damaged. When picking up items outdoors, relative displacement is unlikely to happen.

(70) Further, the full-finger wear-resistant and anti-slip gloves **1020** are provided with a number of friction particles, and the friction particles here may include, but not limited to, spherical shapes, which increase the roughness of the outer surfaces of the full-finger wear-resistant and anti-slip gloves **1020**, thus prevents grabbed items from slipping off the wearer's grip.

(71) Fourth, in this embodiment, the ankle position of the wear-resistant and anti-slip shoe covers **1021**, the wrist and waist positions of the protective clothing are provided with hook and loop fasteners with their surface made of reflective material, so as to ensure the safety of field operations and tighten and fix the protective clothing.

(72) With reference to FIG. **10**, according to embodiment 2, the reusable protective clothing for

epidemic prevention is made of a protective clothing material including a high-density woven fabric layer **30**, a middle-layer fabric **40** and an outer-layer fabric **50** which are designed with a three-layer sandwich structure, and the inner layer is high-density woven fabrics that have good moisture permeability and tear resistance.

(73) It should be noted that, for the sake of conciseness, the implementation principles and technical effects of this embodiment that are not mentioned in this embodiment can be understood by referring to the corresponding content in embodiment 1.

(74) In summary, in combination with embodiment 1 and embodiment 2, the protective clothing of the present disclosure may include a one-piece protective clothing designed for epidemic prevention personnel and volunteers and a medical protective clothing designed for medical personnel. The one-piece protective clothing of the present disclosure can maximize prevention against exposure to pathogenic microorganisms in the wearing and the taking off process, the term “one-piece” reflects in the design that only the nose and mouth are accessible from the outside and the rest of the whole body is fully enclosed to form a closed loop. The protective clothing is lightweight and does not need any additional auxiliary device or means. Also, the protective clothing is able to be put on and taken off as a whole and thus the putting-on and taking-off can both be accomplished by a single act.

(75) The medical protective clothing and the reusable protective clothing for epidemic prevention of the reusable protective clothing are both made of a material designed with a three-layer sandwich structure including a high-density woven fabric layer **30**, a middle-layer fabric **40** and an outer-layer fabric **50**.

(76) Specifically, the material of the inner layer of the medical protective clothing described in embodiment 1 is a high-density woven fabric with good moisture permeability and tear resistance, the middle-layer fabric **40** is PTFE composite membrane that is light, thin, breathable, and can effectively isolate pathogens and withstand a temperature of 260° C. The outer-layer fabric **50** is a cotton cloth treated with chitosan molecules which have an antibacterial functionality and can withstand high-pressure disinfection. The material of the protective clothing is not only lightweight, and wear-resistant, but also has a degree of air permeability, and the fabric can withstand moist heat sterilization, namely, the material can be treated under 115° C.-121° C. for 15 min-30 min, and withstand more than 50 times of moist heat disinfection, so as to realize reuse of the protective clothing for multiple times and reduce the generation of medical waste, thus satisfying the requirement of having the ability to withstand the treatment of autoclave of hospital or laboratory to achieve repeated use.

(77) The protective clothing for epidemic prevention described in embodiment 2 is different from the medical protective clothing described in the embodiment in that their materials have some differences. In embodiment 2, the middle-layer fabric **40** may comprise an organosilicon compound with good water resistance or fluorocarbon compound, and the outer-layer fabric **50** may be wear-resistant and oxidation-resistant polyester fiber, such material can satisfy the requirement of having the ability to withstand oxidative disinfection for repeated use. The material of the protective clothing is not only lightweight and wear-resistant, but also has a degree of air permeability. The protective clothing can withstand immersion in chlorine-containing disinfectant with a concentration of 1000 mg/l for 60 minutes, and more than 50 times of disinfection, thereby realizing reuse for multiple times and reducing the production of medical waste.

(78) The high-density woven fabric mentioned above includes weft yarn of superfine composite fiber composed of 30% nylon and 70% polyester, and warp yarn composed of 70 D/72 ordinary polyester yarn. The woven high-density fabric shrinks by 40% in the weft direction to become a denser fabric, the inter-fiber pore size can be controlled at 7 μm, the monofilament fineness is less than 0.2 D, and the monofilament density can reach 68580/(2.54 cm).sup.2, the moisture vapor permeability is 5000-7000 g/m.sup.2.Math.24 h, the hydrostatic pressure resistance is 700 mm or more, and it has both warmth retention and elasticity properties.

(79) The pathogen barrier effect of PTFE composite membrane described above will not be damaged even after being washed multiple times. As shown in the table below, according to the GB19082-2003 and WSB58-2003 standard, the protective clothing material containing PTFE composite membrane hasn't been washed and the protective clothing material containing PTFE composite membrane after 15 washes are compared by the following properties: filtration efficiency, synthetic blood penetration, airborne microbial barrier performance, liquid poliovirus barrier performance under natural pressure, and liquid poliovirus barrier performance under negative pressure.

(80) TABLE-US-00002 protective GB 19082-2003 WSB 58-2003 clothing Specified Specified Test item material indicators indicators Filtration efficiency (%) unwashed $>95 \geq 70\%$ — 15 washes >95 Synthetic blood unwashed qualified Impermeable — penetration (pressure 15 washes qualified 13.8 kPa) Airborne microbial unwashed >99 — >99 barrier performance (%) 15 washes >99 Liquid poliovirus barrier unwashed 100 — >99 performance (%) (under 15 washes >99 atmospheric pressure) Liquid poliovirus barrier unwashed 100 — >99 performance (%) (under 15 washes >99 negative pressure)

(81) It can be learned from the table that whether the protective clothing material containing PTFE composite membrane hasn't been washed or has been washed 15 times, the properties of filtration efficiency, synthetic blood penetration, airborne microbial barrier performance, liquid poliovirus barrier performance under atmospheric pressure, and liquid poliovirus barrier performance under negative pressure are not changed.

(82) Referring to FIG. 11, as the mass concentration of the chitosan antibacterial agent SA-50DS increases, the rate of bacteriostasis of the protective clothing textile of this embodiment also increases. The reason is that as the mass concentration of chitosan antibacterial agent SA-50DS increases, the amount of antibacterial agent bonded to the textile increases, so the antibacterial performance of the finished textile is better. When the mass concentration of chitosan antibacterial agent SA-50DS reaches 60 g/L, the rate of bacteriostasis of the finished textile is close to 100%.

(83) Specifically, the polyester fiber is a knitted polyester fiber sample. Compared with the woven pure cotton sample, the woven polyester fiber sample and the knitted pure cotton sample, the knitted polyester fiber sample has the best wear resistance and oxidation resistance. The abrasion resistance of woven pure cotton sample, woven polyester fiber sample, knitted pure cotton sample, and knitted polyester fiber sample under dry and wet conditions is shown below.

(84) TABLE-US-00003 Abrasion cycle in Abrasion cycle in Sample dry state wet state Woven pure cotton sample 2050 1850 Woven polyester fiber sample 3900 3500 Knitted pure cotton sample 1850 1500 Knitted polyester fiber sample 3500 3300

(85) It can be learned from the comparison that when compared to the woven pure cotton sample, the woven polyester fiber sample and the knitted pure cotton sample, the knitted polyester fiber sample with the polyester fiber can achieve the best wear resistance and oxidation resistance.

(86) The above description merely includes preferred embodiments of the present disclosure, and is not intended to limit the present disclosure. For those skilled in the art, the present disclosure may have various modifications and changes. Any modification, equivalent replacement, improvement, and so on made within the spirit and principle of the present disclosure shall be included within the scope of protection of the present disclosure.

Claims

1. A reusable protective clothing comprising: a one-piece protective suit including integrated top and trousers with attached gloves and shoe covers; and a protective headpiece having a face shield, wherein a front side of the protective headpiece is provided with a full-vision goggle with 180-degree field of view, the full-vision goggle is integrated with the face shield, a back side of the protective headpiece is provided with a protective hood made of a material of the one-piece

protective suit, the protective hood is integrated with the top, the front side of the protective headpiece is provided with an air vent for a mouth and a nose configured according to facial features of a wearer, the air vent for the mouth and the nose is located at a position corresponding to the mouth and a bridge of the nose of a wearer, and an edge of the air vent is provided with a silicone sealing ring and a filter membrane, the filter membrane is fixed around the air vent via the silicone sealing ring and the silicone sealing ring fits with a shape of the wearer's face without leaving a gap; and an outer side of the protective headpiece is provided with ear-shaped hooks; a width of an inner circumferential portion of the silicone sealing ring proximal to the air vent for the mouth and the nose and being in contact with an inner wall of the protective headpiece is greater than a width of an outer circumferential portion of the silicone sealing ring distal to the air vent for the mouth and the nose and being fit with the wearer's face; and an elastic belt is arranged along an edge around an outer contour of the filter membrane.

2. The reusable protective clothing according to claim 1, wherein the protective headpiece further comprises a latex band arranged inside the protective headpiece at two sides of the air vent for the mouth and the nose to assist the positioning of the air vent for the mouth and the nose.
 3. The reusable protective clothing according to claim 1, wherein the protective headpiece further comprises a support ring arranged at a top part of the protective headpiece.
 4. The reusable protective clothing according to claim 1, wherein the top comprises a back body covering piece and a front body covering piece, the protective hood is connected to a neck position of the back body covering piece and extends to the top part of the protective headpiece, and a first end of a waterproof zipper is arranged at a hip position at a back side of the trousers, a second end of the waterproof zipper extends through the back body covering piece to the top of the protective hood, and the waterproof zipper is covered with a flap.
 5. The reusable protective clothing according to claim 1, wherein the gloves are full-finger wear-resistant anti-slip gloves.
 6. The reusable protective clothing according to claim 1, wherein the gloves are half-finger gloves, and a wrist position of each half-finger glove is provided with nano tape.
 7. The reusable protective clothing according to claim 1, wherein the shoe covers are provided at bottom positions of trouser legs and have wear-resistant and anti-slip property.
-